# Defensive Cybersecurity CITS2006 Portfolio Part 1


## Sam Rose

**24527391**


## Github Repository Link

**Github Link:** [https://github.com/Stegossaurruss/DefensiveCybersecurityPortfolio](https://github.com/Stegossaurruss/DefensiveCybersecurityPortfolio)

## Information

### Github Repo

- My repository, linked above, contains all attachments, extra files, proof or other explanations for the task in Portfolio part 1.
- At any point if an attachment is referenced, it will be present in the relevant Github directory for the task.
- I could have included almost all the proof here, but for a cleaner report the expanded versions all exist in Github, separated by task.

# Self Assessment

## Explanation

- I believe I have completed each task to an outstanding level.
- I have discovered, researched or found real-world examples for each point and provided relevant proof for each.

## Table

| Task Number | Mark out of 2 |
|---|---|
| A1. | 2 |
| A2. | 2 |
| A3. | 2 |
| A4. | 2 |
| A5. | 2 |
| A6. | 2 |
| A7. | 2 |
| A8. | 2 |
| A9. | 2 |
| A10. | 2 |
| A11. | 2 |
| A12. | 2 |
| A13. | 2 |
| A14. | 2 |
| A15. | 2 |
| A16. | 2 |
| A17. | 2 |
| A18. | 2 |
| A19. | 2 |
| A20. | 2 |
| A21. | 2 |
| A23. | 2 |
| A24. | 2 |
| A29. | 2 |
| A30. | 2 |

# A1. Security Concepts on Campus

### Campus Card Readers

- Card readers are used to provide access control to certain buildings, areas or resources on campus, a clear example of something a student could have to verift their identity.
- I discovered this one on the door of the Computer Science building at UWA

### Eduroam Login and Security

- University wifi networks require student ID and password to login, and the data is secured with WPA2/WPA3 protocols. On top of that, there are other security measures on the network such as forced lockout after too much failed password attempts.
- I discovered this by thinking about university services that require a login.

### Security Cameras

- A very straightforward method, security cameras are used to protect sensitive areas and locations, so that suspicious/illegal behaviour can be detected and recordings can be checked in the case of a breach.
- I discovered this camera on the ground floor of Reid Library.

---

# A2. Discover security concepts used in a public space

### Parking meter (combined with parking checkers)

- This parking meter is on the west side of UWA, just out the front of the Computer Science building.
- Parking meters provide access control to parking spots, requirement a payment depending on a set of rules set by the parking space.
- If the payment is not provided, in this case a UWA parking officer will issue a fine to the offending car.
- This puts financial incentive to follow the access control rules set by the University for the parking.

### Traffic cameras

- Traffic cameras are usually paired with traffic lights, and are used to enforce the following of the rules set by the traffic lights.
- If a driver runs a red light or stops too far into the intersection, this traffic camera will record that breach as evidence to issue a fine to the driver.

- It is very dangerous to disobey traffic lights and it is important to the security of all drivers on the road that everyone is following the same rules to ensure safety, and traffic cameras help to enforce this.

### Tag-accessed drop gate

- This is the drop gate used to access the parking lot for the sailing club I am a member of, and use to park and then walk to UWA.
- I have a small blue tag that I press against the reader, which causes the gate to drop to the ground and I can drive over it.
- This is an example of "something I have" access control to a restricted parking lot reserved for members only.

---

## A3. Discover security concepts used in your home

### Motion-based home security cameras

- My house has a few of these set up around it, especially covering all entrances.
- They automatically upload footage to a cloud server when motion is detected.
- I discovered this while checking out the security features of my house.
- Security concept: surveillance

### Ring Home Security Doorbell

- This doorbell has a camera, also designed to sense motion and will also record upon the doorbell being rung.
- It can send an alert to mobile devices, and be configured to alert a user if suspicious motion is detected (at strange times, etc.)
- I discovered this while checking out the security features of my house.
- Security concept: deterance and evidence collection

### Pinpad and biometric door lock

- I left this lock out of my other lists for inclusion here, it has a pinpad based code unlocking system, as well as a fingerprint scanner.
- Each member of my family has a different code, so the user can see who has accessed the door, or whose code was used.
- It has the added bonus of not requiring a key, which means it cannot be misplaced into the hands of a potential threat actor.
- I also discovered this searching for security measures in my home.
- Security concept: access control

---

## A4. Discover a vulnerable website

### Bridge Association of WA

- Website: http://www.bawa.asn.au/index.asp
- This website is vulnerable because it does not use HTTPS encryption, so any information input, sent or received by the site is vulnerable to eavesdropping, interception, modification and other threats. As well as this confidentiality and integrity are broken because your information is accessible by others, and it is possible to modify it through something like a man in the middle attack.
- I discovered this website through searching for vulnerable websites, browsing through lists of clubs and associations in Perth.

---

## A5. Discover cryptographic implementation used online

### Whatsapp End-to-end Encryption

- Whatsapp uses end-to-end encryption on messages sent, meaning even Whatsapp cannot see the content of your message. A message is encrypted on my device and then that is sent to the recipient, whose device can decrypt the message.
- Curve25519 is the public key algorithm used secure communications.
- I discovered this by researching how social media platforms encrypt their data, and then navigating in the Whataspp mobile app.

---

## A6. Discover cryptographic implementation used offline.

### FileVault on Mac

- Apple iOS has the option to enable FileVault, which is a method of encrypting the device's disk.
- It uses AES encryption to encrypt the entire hard drive when the device is locked, leaving it unreadable.
- When the device is unlocked, the password unlocks the decryption key.
- All of this happens while offline.
- I discovered this while researching about offline encryption, and then finding the relevant section in Settings.

---

## A7. Discover cryptography used in modern networks

### WPA2/WPA3 Wifi Encryption

- Many modern wifi networks implement WPA2, and especially WPA3 encryption the secure the network

- It uses AES symmetric encryption and key exchange protocols between devices and routers
- When a device connects, a secure handshake is performed between device and router which allows a shared encryption key to be established and then all data sent over the network is encrypted using AES
- As a note, WPA3 uses stronger, more modern key exchange methods and is more secure against password guessing threats.
- Attached is network information for UWA Unifi (WPA2), and my phone's personal hotspot (WPA3).
- I discovered this while troubleshooting my personal hotspot's connection to my Playstation 4, combined with research about Wifi cryptography.
- Link: https://www.portnox.com/cybersecurity-101/network-security/wpa3/

---

## A8. Discover cryptography used in Internet of Things devices

### Google Home

- I have a Google Home in my place that I use to play music, control lights, and control my TV by voice
- I know it connects to wifi and communicates with my phone and voice, but after some research I discovered that it uses TLS encryption and uses certificates and keys to authenticate the connection with my phone.
- It also encrypts any voice commands (again with TLS) I give it to make sure my privacy is upheld.
- I discovered the cryptography by researching about my Google Home (an IoT device) and how Google uses cryptograph measures to uphold CIA.
- Attached is a photo of my Google Home in my room.

---

## A9. Discover privacy technique used online

### Cookies and Tracking

- Cookies are used to track user data, and websites are required to notify the user of cookies being used to track their data
- Often in the form of a pop-up, there will often be options to reject or accept the use of cookies on the wbesite, or to use the least amount of cookies possible for the site to function.
- This is so that the user can control their level of privacy by restricting how much their acitvity can be tracked.
- I discovered this by learning about cookies in Agile Web Development and doing some research on regulations surrounding cookies.

---

# A10. Discover privacy technique used offline

### Lock Apps behind Face ID on iPhone

- I can choose to require Face ID in order to open a certain app.
- In order to open that app, I must then provide my biometric facial features, and if that fails input my password.
- This check works when my phone is offline, and is a device security measure to ensure privacy, for example if my unlocked phone was stolen that app would still be secure.
- I discovered this by researching iPhone security and locking tools, and also trying not to have duplicate evidence in my portfolio.

---

# A11. Discover 5 unique access control devices

### Barcode

- Here I have a ticket for a festival in my Apple Wallet, and then barcode on it was scanned to allow entry into the festival.
- This is an example of access control using barcodes received upon payment, that permits access to an area for a service.
- I discovered it by buying the ticket and researching forms of access control.

### RFID scan

- Here is another ticket I bought for a different festival, and instead of a barcode the ticket is in the Apple Wallet.
- This ticket uses NFC (Near Field Communication) similar to RFID to be scanned and matched against purchased tickets to permit entry.
- I discovered it again by buying the ticket and scanning it to access the event.

### Location-based blocking

- Some websites, especially video and streaming services restrict some content based on country or location.
- Using a VPN I was watching a trailer only accessible in the USA and Canada, and upon turning off the VPN received this message.
- Access to the service is restricted based on geographic location.

### Captcha

- Captcha is used to ensure that a human being is interacting with a site and not a bot.
- There are many different forms of captcha, but ReCaptcha is a very common protocol used.

- It restricts access to website and other services such as login by ensuring that only human users are permitted access.

### Netflix device access control

- Netflix lets you control which devices have access to your account, and which devices can remain logged in.
- I was able to access these controls from within my Netflix account, and logout and restrict access to the account from certain devices.
- I discovered this by researching parental controls in Netflix for this portfolio and instead finding these access controls.

---

## A12. Discover 5 unique offline security tools

### Safe

- Safes are used to provide physical access control to items or physical information.
- With a code is ensures only authorised individuals have access to what is inside, preventing theft and ensuring the security of its contents.
- I discovered this safe at my job, where only the site manager knows the code.

### Security Camera

- Security cameras are a very commonplace security measure to ensure that sensitive locations are watched at all times, even if not actively.
- This one is in the 3rd floor of Reid Library, watching me complete my portofolio and ensure I'm not threatening the university's resources.

### Physical Locks

- Physical locks are an extremely common offline security tool used to restrict access to certain areas. There is another portfolio point about locks, so this one is unique to demonstrate their role as security tools.
- Depending on the lock, one might need a key, code or biometric to be authorised and gain access to an area behind a lock.
- This one is also on the 3rd floor of Reid Library, and I have absolutely no idea what is behind it, so it is doing it's job well.

### Smoke Detector

- Smoke detectors are an offline security measure that detect smoke and other signs of fire in an area. They may be connected to further systems by a network, especially in a university, but the detector itself has an alarm that will sound to alert inhabitants of danger.

- For this reason they are often battery powered, so that if there are network or power issues they are not affected.

### Signage

- Signs or warnings with information regarding the item, area or object they concern are also common security measures.
- While simple and not always effective, they indicate that only authorised individuals may access an area or that there is danger ahead.
- This information supports security policies in place and informs users of them.
- Here is the door to the roof of Reid Library, full of warnings about danger of RF levels and indicating the only authorised people may enter.

---

## A13. Discover 5 unique online security tools

### Password Manager

- Password managers can be used to store many passwords, so that each password can be strong and unique, not overly relying on memorising all of one's passwords.
- I used MSecure, the login page below, and it is extremely useful in encouraging me to use diverse and complex passwords.

### Browser Security Warning

- Most browsers have security warnings if they detect that a site is not secure. This is a valuable tool for ensuring a user is aware not to share sensitive information or fall victim to other potential threats.
- I discovered this through my research for the group project for this unit, and here is a warning pop-up prompted by an HTTP only site.

### Email Spam Folder

- Spam filters in email systems automatically detect and isolate malicious or suspicious emails such as scams or malware.
- While not perfectly effective, it is an overlooked and important tool for protect users from these threats.
- I discovered this through some research and my existing knowledge of online security.
- My spam folder, full of scam emails from just this month.

### Personal Access Tokens

- A Personal Access Token is used instead of a password to authenticate login, in my case Github login

- I discovered this because my Github username and password weren't working properly for a shared repository.
- Using a PAT instead of password reduces my password exposure and solves a problem, but when used in other cases they are excellent tools for fine-grain, personalised access control and can be very useful security tool in this way.

### Browser Extension Security Tools

- Another underrated security tool I use is UBlock Origin, as it not only blocks ads but also malicious scripts and trackers on webpages.
- While it's primary function is to block ads for me, other browser extensions like Malwarebytes Browser Guard and Bitwarden have additional uses protecting browser usage from threats.
- I discovered UBlock through wanting to block ads, but after some more research I came across more specific browser security extensions.

---

## A14. Discover 5 AI-enabled security solutions

### AI Fraud Detection

- AI based fraud detection has been implemented by many businesses like Paypal to analyse transaction data.
- Location, patterns of spending and behaviour of each user is tracked to identify if suspicious activity is detected.
- AI is used to learn what a normal purchase or transaction looks like and then flag and prevent unusual ones.
- I discovered this by researching the uses of AI in defensive cybersecurity
- Link: IBM (https://www.ibm.com/topics/fraud-detection)

### Neural Networks with Facial Recognition

- Face ID and other facial recognition programs are commonly used as a security measure now, like using a facial scan to unlock a device.
- Many of these programs, such as Face ID, are powered by neural networks, an AI solution to security challenges like passwords.
- I discovered the use of neural networks through my knowledge of facial recognition and some confirming research.
- Link: Microsoft Azure (https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-face-recognition#:~:text=Face%20recognition%20is%20a%20form,identify%20or%20group%20a%20face.)
- Link: Apple (https://support.apple.com/en-in/102381)

### AI Network Threat Detection - Darktrace

- Darktrace is cybersecurity tool that uses AI to detect unusual behaviour in a network.
- It does this by first learning what normal activity on the network looks like, then identifies anomalies.
- Because it has an image of "normal" behaviour, it can also detect zero-day exploits, very difficult to do without AI models.
- I discovered it by researching prominent and effective AI security tools in 2026.
- Link: Darktrace (https://www.darktrace.com/products/network)

### AI System Protection - Cloudstrike Falcon

- Cloudstrike Falcon is another cybersecurity tool that focuses on protecting devices like computers and servers from threats.
- Like Darktrace, it monitors system activity and implements AI to identify suspicious or malicious behaviours.
- Depending on what platform and service is provided, it can also prevent threatening actions from being undertaken before they cause damage to the system it is protecting
- I discovered it again by research effective AI security tools.
- Link: Cloustrike Falcon (https://www.crowdstrike.com/en-us/platform/)

### Automated Content Detection - Youtube

- Youtube uses automated content detection to indentify content that goes against their content policies and terms of service.
- This helps to better enforce their rules on a huge scale, and protect users from harmful, threatening or dangerous content.
- This applies to both videos and comments, and greatly helps to improve security on a platform as big as Youtube
- I discovered this through more research of AI-based security implementations, as I don't always see proof of AI usage on the surface.
- Link: Youtube (https://www.youtube.com/howyoutubeworks/our-policies/#:~:text=How%20Community%20&%20Ad%2DFriendly%20Guidelines,they're%20advertising%20on%20YouTube.)

---

## A15. Discover 5 recent security incidents

### Discovery

- All of these incidents were found by browsing recent incidents, alerts and advisory boards in Australia, and around the world, and all take place in the last year.

### Victorian Department of Education

- The organisation was hit by a major data breach that affected all of the government schools in Victoria.
- Attackers accessed personal information and passwords of past and present students.
- A clear breach of the confidentiality of all students involved
- Link: (https://www.cybernewscentre.com/15th-january-2026-cyber-update-victorian-schools-hit-by-major-data-breach/?ref=daily-cyber-update-newsletter)

### Smile Orthodontics

- The business was targeted by hacker group SafePay, and much data was accessed and leaked from their systems.
- Data includes home addresses of staff, personal emails, medical certificates, payment plans and patient treatment histories.
- This data leak is another clear example of breach of confidentiality, with very sensitive information becoming available publically.
- Link: (https://www.cyberdaily.au/security/13304-exclusive-safepay-ransomware-claims-hack-of-smile-team-orthodontics?utm_source=newsletter&utm_campaign=Daily&utm_medium=email&utm_content=2026-03-11&utm_emailID=fe740e8223e19c88ab5a9844d8dad2669ae410a794b23cad1d082072530b5507)

### Prosura

- A rental car insurance company, Prosura suffered a major data breach, and threat actors are selling the data on public forums.
- Data accessed includes names, contact details, policy information, travel destinations, drivers' licences of around 3000000 customers.
- Reportedly, Prosura did not meet the ransom demands of the attackers and the attackers started publishing the data.
- This is another clear example of financially motivated ransomware attacks that undermine the security of the Australian population.
- Link: (https://www.cybernewscentre.com/14-january-2026-prosura-breach-exposes-300k-customers/?ref=daily-cyber-update-newsletter)

### Qantas

- In mid 2025, Qantas was targeted by hacker group Scattered Lapsus$ Hunters, and a breach occured through a 3rd party call centre.
- While more sensitive data like financial details and identity documents was not accessed, names, dates of birth and frequent flyer points were all subject to the breach.

- Working with groups like the Australia Federal Police to resolve it, this was another huge-scale breach of confidentiality.
- Link: (https://australiancybersecuritymagazine.com.au/stolen-qantas-customer-records-surface-on-dark-web/)

### University of Mississippi Medical Centre

- THe UMMC was subjected to a ransomware attack and as a result, had to shut down clinic locations in Mississippi.
- The attack took down their IT systems and blocked access to their system of medical records, rendering everyday hospital services unavailable. Essential services were still active, but it took 9 days to regain use of their systems and longer to resolve the attack.
- This time ransomware was used to hinder the availability of the systems and data, putting lives in danger and leaving the 10000 UMMC employees without work for the duration.
- Link (https://www.bleepingcomputer.com/news/security/university-of-mississippi-medical-center-closes-clinics-after-ransomware-attack/)

---

## A16. Discover 3 local security incidents

### Legal Practice Board of Western Australia

- Through some research about cybersecurity incidents in Perth, I discovered the the LPBWA was attacked and breached in mid 2025.
- The Direwolf ransomware group accessed some corporate data and some personal information, but I was not able to find technical specfications due to a legal injunction I believe.
- Direwolf ransomed the data and compromised the security of the LPMWA, who are now vigilant for misuse of that data.
- Link: (https://www.lpbwa.org.au/cyber-incident)
- Link: (https://www.cyberdaily.au/security/12158-exclusive-legal-practice-board-of-western-australia-confirms-dire-wolf-ransomware-attack)

### UWA cyberattack

- UWA faced a cyberattack in mid 2025, with the university detecting unauthorised access to university password information.
- As a result of this breach, all staff and students were locked out of all UWA systems and were required to change their passwords.
- It was not disclosed how many students' passwords were accessed, but UWA immediately began enacting counter-measures upon attack.

- I discovered this attack because of the communciations to students last semester. (Also had to change my password.)
- Link: (https://www.cybersecurityintelligence.com/blog/university-of-western-australia-forced-into-online-lockdown-8638.html)

### Harry Perkins Institute

- A ransomware group known as Medusa successfully accessed and stole a large amount of data, around 4.6 terabytes.
- This data largely consisted of security camera footage throughout the facility as well as some other information.
- A ransom of $700000 was posted and then ignored, and the attackers claimed to have posted the data online.
- I discovered this with more research of cybersecurity incidents in Perth, a clear breach of confidentiality.
- Link: (https://www.watoday.com.au/national/western-australia/perth-medical-research-facility-internal-camera-footage-published-online-hackers-claim-20240716-p5ju9f.html)

---

## A17. Discover 10 different types of locks in use

### Discovery

- I found all of these locks in various places around Perth, and one online login page (I ran out of locks)

#### 1. Biometric fingerprint scanner

- Used with a thumb to unlock this model of phone

#### 2. Safe numbered keypad

- Number-pad latch used to protect the contents of a safe

#### 3. Key-operated padlock

- Used with a key to secure entry to an area

#### 4. Car interior door lock

- Used to lock the car from the inside, to prevent outside access

#### 5. iPhone passcode

- Pinpad used to control access to the phone

### 6. Colour-accessible bathroom lock

- Used to lock a bathroom and indicate occupancy of said bathroom

### 7. Code-operated padlock

- Used with a code to restrict access to an area

### 8. Pinpad-operated door lock

- Used to provide access to this hallway for those with authentication (knowing the code)

### 9. Interior deadbolt lock, no key

- Simple lock with a swinging deadbolt that can only be opened from inside

### 10. Online login page

- Login page used to control access to features that might require an account.

---

## A18. Discover two hallucination cases when using a generative AI system

### Fake legal case citations

- Through researching prominent AI hallucination cases, I found a U.S. court case (Mata v. Avianca, 2023)
- A lawyer used ChatGPT to generate legal arguments and it produced completely fake case citations that did not exist.
- The AI generated plausible but entirely false information, and when confronted to verify the validity of the cases, ChatGPT confidently insisted that the cases existed, when in fact they did not.
- A clear hallucination case in which ChatGPT confidently provided the user with false, made-up information.
- Link: (https://www.bbc.com/news/world-us-canada-65735769)

### Google bard incorrect factual information

- Upon the debut of Google's Bard in early 2023, an LLM chatbot, it hallucinated information about the James Webb Telescope.
- It claimed that the telescope took the first ever image of a planet outside of our solar system.
- This is a false claim, as the first photos of an exoplanet were taken in 2004, 14 years before the telescope launched, as evidenced on NASA's website.

- This is a significant hallucination case because it was the first demo of Google's - a reputable tech company - chatbot model, and immediately encountered errors with the AI making up information.
- Link: (https://www.theverge.com/2023/2/8/23590864/google-ai-chatbot-bard-mistake-error-exoplanet-demo)
- Link: (https://science.nasa.gov/exoplanets/)

---

# A19. Join a CS/DS/cybersecurity club

### UWA Data Science Club

- Joined the UWA Data Science Club

---

# A20. Participate in a discussion with your friends about cybersecurity events

## Our discussion

- We held a discussion in the Week 6 Lab to further our cybersecurity knowledge and fulfill the requirements for this task.
- Here are a number of topics we discussed, mostly focused around the rise of AI and it's relevance in cyberspace.
- Attached is photo of us in the lab after our chat.

## Emerging cyber threats with AI

### Prompt Injection

- Most of the members of the discussion had done some research on prompt injection for task A22, and as a result we'd learnt about the prominence of prompt injection threats to large language models.
- One member metioned that prompt injection attacks are the numer one risk in the OWASP for LLMs and prompt injection vulnerabilities are present in over half of controlled AI systems.
- While not all of us had success in performing a prompt injection thus far, we all agreed that we'd encountered similar problems with AI, like in Lab Quiz 1 in which the PDF providing had prompts inject to mislead the user and make the challenge more difficult without strategic use of the LLM.

### Social Engineering

- Another point of discussion was the interaction of AI with social engineering, and a few good points were raised.

- Tools like voice replicators and tone replicators can be used to very effectively fool victims into being exploited.
- As AI systems are used to enhance security, like AI inbox filtering with Gmail, users may become more complacent, less aware, and as a result possibly be more susceptible to phishing or other similar scams.
- On top of this, the ease of use and volume capabilities of LLMs make it significantly easier to create large scale, convincing attacks from a social engineering perspective, lowering the skill, knowledge and effort required for successful malicious activities.

### AI data collection

- Another point of concern raised was with the way the LLMs gather data on users. Usage over time provides the model with a surprisingly large amount of personal information, especially if the user isn't careful to restrict what it tells the model.
- The group raised concerns over how that information is stored, what it is used for, and possibly whether or not that data, and all user data is being sold to third-party data buyers.
- We weren't sure exactly about the policies involved, legally and with individual LLMs, but with our general distrust for large companies we all agreed that we should learn more about it.

---

## A21. Participate in an online cybersecurity discussion

### TryHackMe Discord

- A community for cybersecurity discussions online, as well as CTF and other challenges.
- I read through the chat in the general channel and there was some discussion over benefits and weaknesses of operating systems.
- I'm looking at buying a new device because mine is quite underpowered and on iOS, I asked the questions below and got some answers from a moderator of the server, suggesting windows over Linux.
- Conversation is below, as well as a screenshot of the whole `#general` channel

---

## A23. Enhance the cybersecurity in your home

### Password Manager

- My sister is a bit notorious for using the same password for everything, so I asked her if she would install a password manager.
- I did this with the goal of encouraging her to using varied, complex passwords for different applications, especially for important accounts that contain sensitive information like myGov, bank, Medicare and university accounts.

- I cannot confirm whether or not she will use the manager to actually implement diverse passwords, but I will continue to encourage her to enhance her cybersecurity measures, as well as use multi-factor authentication.
- In the attachment I have provided a photo of her logged into the password manager on her laptop (Account shown in the menu on the top left).

---

# A24. Teach your family about cybersecurity topic of your choice

## Teaching my dad about RSA key-pair encryption

### Overview

- I made a brief, simple powerpoint presentation on asymmetric encryption to teach my dad.
- I chose this topic because he's quite technologically inclined and has a fair existing knowledge of basic cybersecurity concepts.
- He's also pretty good at math, so understanding moduli is not too difficult.

### Question

- He asked a question about just how long a 1024bit number is in digits, for a real world comparison, and we found out that its 309 digits.
- With some more research, we found out that with RSA-1024, there are around 2.8*10^147 primes that could be chosen for p and q.
- This is measurably more than atoms in the observable universe, so the chance of picking the same prime is mathematically impossible.
- We also found that in order to store all primes 1024bits and under, it would require around 50000 trillion times more computer storage than currently exists on Earth.

### Attachments

- Find the powerpoint in this folder, and a photo of my dad and his signature for proof!

---

# A29. Find a publicly available AI-generated image, video, or audio clip, use at least one detection or verification tool to analyse it

### AI image detection

- I found an AI-generated image of a woman at https://generated.photos/face/joyful-white-young-adult-female-with-medium-brown-hair-and-brown-eyes--5e6804fe6d3b380006d48c27

- I then went to https://truthscan.com/account and uploaded the image.
- Unsurprisingly, the image was found to be synthetic with a 97% certainty, confirming its generation by AI.

---

# A30. Complete an online cybersecurity module

### Codecademy - Introduction To Ethical Hacking Course

- Completed the Codecademy Introduction to Ethical Hacking Course.
- Certificate is attached.

---